

ENN584 Robot Systems

Assessed Lab2: Occupancy Mapping & Sensor Fusion

Anthony Vanderkop and Thierry Peynot

Sensor Data Fusion

This lab practical builds upon the occupancy grid mapping implementation that you initiated in Prac. 5. Please refer to Prac 5 for additional details on this section of the work.

Your task in this practical is to implement the incomplete methods in the Python template `sensor_fusion.py` to perform occupancy grid mapping with a robot in a 2D environment equipped with:

1. a laser scanner only,
2. a radar scanner only,
3. both, with early sensor fusion methods,
4. both, with late sensor fusion methods.

For the purpose of this exercise, we consider that the laser scanner is more accurate than the radar scanner (which is usually the case) but has shorter range and is vulnerable to obscurants in the environment, such as smoke or airborne dust (in this practical we will use a simulated version of smoke particles).

Read through the code in both `sensor_fusion.py` and in the `util_funcs.py` script to understand the tools you have available to you. However, you should only edit code in the main script `sensor_fusion.py` (i.e. do not modify the content of `util_funcs.py`).

You are free to edit the code in this script in any way you like, the code provided is merely a suggestion and skeleton for you to build on if you would like.

For the assessment in the Week 11 session you will need to demonstrate the following.

1. Functional occupancy mapping with each sensor individually
2. Understanding of what early and late sensor fusion techniques mean and how they differentiate from each other. Where are they both useful?
3. Attempts at implementing and testing sensor fusion to improve robustness under uncertainty, for the purpose of occupancy grid mapping.
4. Understanding of how the code works, how changing certain parameters will affect performance, and of other core concepts behind occupancy mapping.

You will demonstrate these items during the prac session by:

- demonstrating the outputs of your code as it runs (print statements, plots, graphs),
- inspecting variables and methods in a live python environment, and through the outputs of your code as it runs (print statements, plots, graphs),
- answering questions from the tutor or lecturer.